

# BankMathBench Evaluation Assumptions and Formula Reference

---

This document summarizes the calculation assumptions and standards used in BankMathBench evaluation in order to minimize result mismatches caused by differences in detailed calculation methods.

## Common Standards

### 1. Time Units

- Time units may be converted across years, months, and days.
- Conversion between annual rates, monthly rates, and daily prorated calculations is allowed.
- For day-based calculations, 1 year is treated as 365 days.
- When needed, 1 month may be treated as 30 days and 1 year as 52 weeks.

### 2. Installment Savings

- Assume payment is made at the beginning of each payment period.

### 3. Loans

- For each repayment method, apply the formulas defined below.

### 4. Rounding

- Dataset answers are rounded to the second decimal place.

## Product-Specific Formulas

### 1. Deposits

Example: the period unit is year.

#### Simple Interest

$\text{Interest} = \text{Principal} \times \text{Annual interest rate (\%)} \div 100 \times (\text{Period in days} \div 365)$

$\text{Early withdrawal interest} = \text{Principal} \times \text{Early withdrawal rate} \div 100 \times (\text{Actual deposit period})$

÷ 365)

## Compound Interest

Maturity amount = Principal × (1 + Annual interest rate (%) ÷ 100) ^ (Period in days ÷ 365)

Interest = Maturity amount – Principal

Early withdrawal amount = Principal × (1 + Early withdrawal rate ÷ 100) ^ (Actual deposit period ÷ 365)

Early withdrawal interest = Early withdrawal amount – Principal

## 2. Installment Savings

For regular deposits of the same amount at fixed intervals.

Example: the payment cycle is monthly.

### Simple Interest

Interest = Monthly deposit amount × Number of deposit months × (Number of deposit months + 1) ÷ 2 × (Annual interest rate ÷ 12 ÷ 100)

Early withdrawal interest = Monthly deposit amount × Actual number of deposit months × (Actual number of deposit months + 1) ÷ 2 × (Early withdrawal rate ÷ 12 ÷ 100)

## Compound Interest

Maturity amount =  $\sum$  [Monthly deposit amount × (1 + Annual interest rate ÷ 12 ÷ 100) ^ (Number of deposit months – i)] (i = 0 to n–1)

Total deposit amount = Monthly deposit amount × Number of deposit months

Interest = Maturity amount – Total deposit amount

Early withdrawal amount =  $\sum$  [Monthly deposit amount × (1 + Early withdrawal rate ÷ 12 ÷ 100) ^ (Actual number of deposit months – i)] (i = 0 to n–1)

Early withdrawal interest = Early withdrawal amount – Total deposit amount

## 3. Loans

### Bullet Repayment

The principal is repaid in full at maturity, and interest is calculated using simple interest and paid monthly.

Monthly interest payment = Loan principal × Annual interest rate ÷ 100 ÷ 12

Total interest = Monthly interest payment × Loan term (months)

Total repayment amount = Loan principal + Total interest

## Equal Principal Repayment

The same principal is repaid every month, and interest is paid based on the remaining principal balance.

Monthly principal = Loan principal  $\div$  Total number of months  
Interest for the month = Remaining principal  $\times$  Annual interest rate  $\div 100 \div 12$   
Monthly repayment = Monthly principal + Interest for the month  
Total repayment amount =  $\Sigma$  (Monthly repayment)

## Equal Principal and Interest Repayment

A compound-interest-based method in which the same amount (principal + interest) is repaid every month.

Monthly repayment = Loan principal  $\times [r(1+r)^n] \div [(1+r)^n - 1]$   
 $r$  = Monthly interest rate (Annual interest rate  $\div 12 \div 100$ )  
 $n$  = Total number of repayment months  
Total repayment amount = Monthly repayment  $\times n$

## Early Repayment

### 1. Early Repayment Fee

Early repayment fee = Early repayment amount  $\times$  Early repayment fee rate  $\times$  (Remaining loan days  $\div$  Loan term)

### 2. Interest After Early Repayment

- Apply a new interest rate to the remaining loan balance, or keep the original contracted interest rate.
- The method for calculating remaining principal and interest depends on the repayment method.

### 3. Total Cost Over the Full Period

Total cost = (Interest paid before early repayment + Interest to be paid after early repayment)  
+ Early repayment fee + Principal paid over the entire period